in Chapter 21, but for the purposes of *Interstellar*, the bulk has just one extra space dimension: four space dimensions in all.

Now, it's very hard for humans to visualize our three-dimensional universe, our full brane, living and bending in a four-dimensional bulk. So throughout this book I draw pictures of our brane and bulk with one dimension removed, as I did in Figure 4.4.

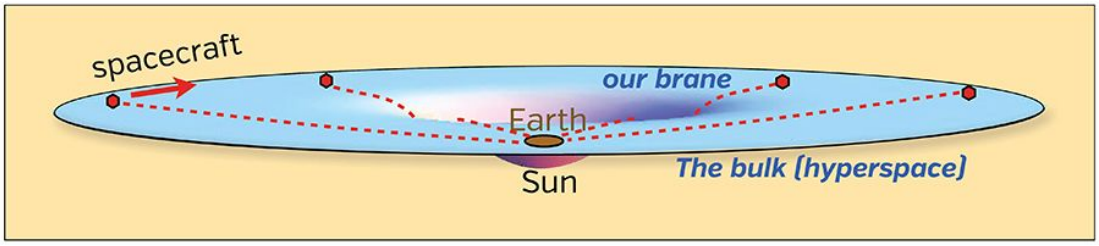


Fig. 4.4. Paths of *Viking* radio signals through the Sun's warped equatorial plane.

In *Interstellar*, the characters often refer to *five* dimensions. Three are the space dimensions of our own universe or brane (east-west, north-south, up-down). The fourth is time, and the fifth is the bulk's extra space dimension.

Does the bulk really exist? Is there truly a fifth dimension, and maybe even more, that humans have never experienced? Very likely yes. We'll explore this in Chapter 21.

The warping of space (warping of our brane) plays a huge role in *Interstellar.* For example, it is crucial to the very existence of the wormhole connecting our solar system to the far reaches of the universe, where Gargantua lives. And it distorts the sky around the wormhole and around the black hole Gargantua; this is the gravitational lensing we met in Figure 3.3.

Figure 4.5 is an extreme example of space warps. It is a fanciful drawing by my artist friend Lia Halloran, depicting a hypothetical region of our universe that contains large numbers of wormholes (Chapter 14) and black holes (Chapter 5) extending outward from our